Semiclassical entanglement of two coupled quantum rotors: Bessel ladders, the separatrix, and phase-space distributions

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Abstract

For two rotors coupled by $\lambda\cos(\theta_1-\theta_2)$ we compare the entanglement entropy of highly excited eigenstates with the entropy of the reduced classical microcanonical distribution, following the programme of Duan, Kovner and Skokov for coupled oscillators, and we ask what changes when the classical distribution is replaced by a quantum phase-space distribution. Every sector of fixed total angular momentum is exactly a quantum pendulum (Mathieu problem), and because the total angular momentum is conserved the reduced density matrix is diagonal in the angular-momentum basis; the phase-space cell is therefore fixed at h and the comparison has no free constant. Deep in the rotation regime the Schmidt weights are $J_j^2(\gamma)$, $\gamma=\lambda I/\hbar^2 n$, and the classical weights are the arcsine law; a momentum-space WKB argument shows that the quantum entropy lies below the classical one by $1-\ln 2$, approached as $\gamma^{-1/3}$, and we confirm this to $\gamma=10^6$. Both entropies are maximal near the separatrix, where the support of the classical momentum distribution has a cusp, and in deep libration $S_0=\frac{1}{4}\ln(\lambda I/\hbar^2)+\frac{1}{2}\ln(\pi e/\sqrt{2})-[8\sqrt{2\lambda I/\hbar^2}\,]^{-1}+O(\hbar^2/\lambda I)$. Replacing the classical distribution by the Wigner function reproduces the von Neumann entropy identically, for every state and coupling: the Wigner function of the reduced state is the marginal of the two-rotor Wigner function and equals the Schmidt weights divided by 2π. The constant $1-\ln 2$ is the entropy carried by the chord (interference) terms of the Wigner function, which survive the marginalisation. With the Husimi function of the reduced state, the Wehrl entropy exceeds the von Neumann entropy by $(1-\ln 2)-D(\sigma)$, with D given in closed form and $D(\sigma)\simeq 0.0802/\sigma$ for a coherent-state width $\sigma\gtrsim 1$, independently of γ. Quantum and classical distributions smoothed by the same coherent state then agree to $O(1/\sigma)$ — at $\lambda I/\hbar^2=200$ and 800, away from the separatrix, to 0.04 for $\sigma=1$ and 0.015 for $\sigma=2$ — but both entropies depend on σ, which takes over the role of the arbitrary cell in the oscillator problem.

Keywords: entanglement entropy, quantum rotors, quantum pendulum, Wigner function, Husimi function, Wehrl entropy, classical–quantum correspondence, separatrix

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1. Introduction

Duan, Kovner and Skokov (DKS) [1] asked how much of the entanglement of a highly excited eigenstate is reproduced by a purely classical calculation. For two coupled harmonic oscillators,

$$H_c = \tfrac{1}{2}\left(p_x^2 + p_y^2 + \omega^2 x^2 + \Omega^2 y^2 + 2Cxy\right), \qquad C \ll \omega^2 \ll \Omega^2, \tag{1}$$

they integrated the microcanonical distribution over the second oscillator, computed the Boltzmann entropy of the reduced distribution $W(x, p_x)$, and compared it with the von Neumann entropy of the reduced density matrix of a WKB eigenstate. The two agree,

$$S_{\mathrm{CE}} = S_E = \ln\left[\frac{\pi C}{\Omega}\frac{\sqrt{E_1 E_2}}{\hbar\omega\Omega}\right], \tag{2}$$

provided the elementary phase-space cell used to make W dimensionless is $\Delta = h/2$; any other cell shifts the classical entropy by a constant. The question is motivated by high-energy scattering, where the entanglement of the probed degrees of freedom has been argued to become the Boltzmann entropy of the final state [2, 3, 4].

The oscillator pair is Gaussian, has a non-compact phase space (so the cell is not fixed by the theory), and has no separatrix. Two rotors on a circle,

$$H = \frac{p_1^2}{2I} + \frac{p_2^2}{2I} + \lambda\cos\left(\theta_1 - \theta_2\right), \tag{3}$$

remove all three restrictions at once while remaining solvable: the angular momenta are quantised, which fixes the cell at h; the coupling is a cosine; and the relative motion is a pendulum with a separatrix. In this paper we treat the semiclassical regime, $\lambda I/\hbar^2 \gg 1$. The weak-coupling regime, where the entropy is obtained in perturbation theory, is treated separately [5]; here it appears only as the small-γ limit of an exact solution (Sec. 4).

A natural alternative to the classical distribution is a quantum quasi-probability distribution on phase space — the Wigner function [6] or the Husimi function [7] — and information measures built on these are an active subject in their own right [8, 9, 10]. One may ask whether the DKS construction works better, or differently, if the microcanonical distribution is replaced by the Wigner or Husimi function of the eigenstate. The rotor pair answers this question exactly, and the answer explains where the classical–quantum difference comes from.

The results are the following.

1. Each sector of fixed total angular momentum is exactly a Mathieu problem, and the entanglement entropy is the Shannon entropy of the Fourier coefficients of a Mathieu function (Sec. 2). The reduced classical distribution is obtained in closed form with an elliptic-integral normalisation and a fixed cell h (Sec. 3).
2. In deep rotation the Schmidt weights are squared Bessel functions and the classical weights the arcsine law; the quantum entropy lies below the classical one by $1 - \ln 2$ in the semiclassical limit (Sec. 4).
3. At generic energies the deficit grows towards the same constant; both entropies are maximal near the separatrix (Sec. 5).
4. In deep libration we obtain the ground-state entropy including the first anharmonic correction (Sec. 6).

5. The Wigner function of the reduced state is the marginal of the two-rotor Wigner function and its entropy is the von Neumann entropy, identically. The constant $1-\ln 2$ is the entropy of the chord terms of the Wigner function. The Husimi (Wehrl) entropy removes a fraction $1-D(\sigma)/(1-\ln 2)$ of the deficit, with D in closed form, at the price of a coherent-state width σ that plays the role of the DKS cell (Sec. 7).

All calculations are given in full; standard integrals are collected in Appendix A and every analytic statement has been checked numerically (Appendix B).

2. The model and its exact reduction

2.1. Basis and conservation of total angular momentum

Each rotor is described by an angle $\theta_i \in [0, 2\pi)$ with $p_i = -i\hbar\,\partial/\partial\theta_i$. Single-valuedness of $e^{im\theta}$ requires $m \in \mathbb{Z}$, so $p_i = \hbar m_i$, and the uncoupled basis is

$$\langle\theta_1,\theta_2|m_1,m_2\rangle = \frac{1}{2\pi}\,e^{im_1\theta_1}e^{im_2\theta_2}, \qquad H_0\,|m_1,m_2\rangle = \frac{\hbar^2}{2I}\left(m_1^2+m_2^2\right)|m_1,m_2\rangle\,. \tag{4}$$

Since $e^{\pm i\theta}\,|m\rangle = |m\pm 1\rangle$ and $\cos(\theta_1-\theta_2) = \frac{1}{2}[e^{i\theta_1}e^{-i\theta_2}+\text{h.c.}]$,

$$\langle m_1',m_2'|\cos(\theta_1-\theta_2)\,|m_1,m_2\rangle = \frac{1}{2}\left[\delta_{m_1',m_1+1}\delta_{m_2',m_2-1}+\delta_{m_1',m_1-1}\delta_{m_2',m_2+1}\right]: \tag{5}$$

the coupling transfers one unit of angular momentum between the rotors. The total angular momentum $J = p_1+p_2$ commutes with H, because $(\partial_{\theta_1}+\partial_{\theta_2})\cos(\theta_1-\theta_2) = 0$, and its eigenvalue $\hbar M$, $M = m_1+m_2$, labels invariant sectors $\mathcal{H}_M$. Within a sector we use the relative label

$$k = \frac{m_1-m_2}{2} = m_1-\frac{M}{2}, \qquad m_1 = \frac{M}{2}+k, \quad m_2 = \frac{M}{2}-k, \tag{6}$$

which runs over $\mathbb{Z}$ for even M and over $\mathbb{Z}+\frac{1}{2}$ for odd M. It is also convenient to write $n = 2k = m_1-m_2$ and to introduce the dimensionless coupling

$$\alpha = \frac{\lambda I}{2\hbar^2}. \tag{7}$$

2.2. Centre-of-mass and relative variables

Classically, introduce

$$\Theta = \frac{\theta_1+\theta_2}{2}, \qquad \varphi = \theta_1-\theta_2, \qquad J = p_1+p_2, \qquad p = \frac{p_1-p_2}{2}. \tag{8}$$

The transformation is canonical, since $p_1\dot\theta_1+p_2\dot\theta_2 = J\dot\Theta+p\dot\varphi$. With $p_1 = J/2+p$, $p_2 = J/2-p$ one has $p_1^2+p_2^2 = J^2/2+2p^2$ and

$$H = \frac{J^2}{4I}+\underbrace{\frac{p^2}{I}+\lambda\cos\varphi}_{H_{\rm rel}}\,. \tag{9}$$

The relative motion is a pendulum with moment of inertia $\mu = I/2$. Its potential has a minimum $-\lambda$ at $\varphi = \pi$ and a maximum $+\lambda$ at $\varphi = 0$, so the separatrix is at

$$E_{\rm rel} \equiv E-\frac{J^2}{4I} = \lambda. \tag{10}$$

For $|E_{\rm rel}| < \lambda$ the relative angle librates about $\varphi = \pi$; for $E_{\rm rel} > \lambda$ it circulates. Quantum mechanically $J = \hbar M$ and $p = \hbar k$.

2.3. Each sector is a quantum pendulum

Completing the square, $m_1^2 + m_2^2 = M^2/2 + 2k^2$, so by (4) and (5)

$$H\big|_{\mathcal{H}_M} = \frac{\hbar^2 M^2}{4I} + h, \qquad h_{kk'} = \frac{\hbar^2}{I} k^2\, \delta_{kk'} + \frac{\lambda}{2}\left(\delta_{k',k+1} + \delta_{k',k-1}\right), \tag{11}$$

a real symmetric tridiagonal matrix. In sector M the wave function factorises, $e^{im_1\theta_1 + im_2\theta_2} = e^{iM\Theta} e^{ik\varphi}$, so $\Psi = e^{iM\Theta} u(\varphi)/\sqrt{2\pi}$ with $u(\varphi) = \sum_k c_k\, \chi_k(\varphi)$, $\chi_k = e^{ik\varphi}/\sqrt{2\pi}$. The operator

$$h = -\frac{\hbar^2}{I}\frac{\mathrm{d}^2}{\mathrm{d}\varphi^2} + \lambda \cos\varphi \tag{12}$$

has exactly the matrix (11) in the basis χ_k, since $\lambda\cos\varphi\,\chi_k = \frac{\lambda}{2}(\chi_{k+1} + \chi_{k-1})$. Each sector is therefore the quantum pendulum H_{rel} of (9), with periodic boundary conditions for even M and antiperiodic ones for odd M. With $\varphi = 2z$ the eigenvalue equation $hu = E_{\mathrm{rel}} u$ becomes the Mathieu equation [11, 12]

$$\frac{\mathrm{d}^2 u}{\mathrm{d}z^2} + (a - 2q\cos 2z)\, u = 0, \qquad a = \frac{4I E_{\mathrm{rel}}}{\hbar^2}, \qquad q = \frac{2\lambda I}{\hbar^2} = 4\alpha, \tag{13}$$

and

$$E = \frac{\hbar^2 M^2}{4I} + \frac{\hbar^2}{4I} \times \begin{cases} a_0 < b_2 < a_2 < b_4 < \ldots, & M \text{ even}, \\ b_1 < a_1 < b_3 < a_3 < \ldots, & M \text{ odd}, \end{cases} \tag{14}$$

which we have checked to all printed digits against exact diagonalisation (Table B.3). The separatrix is at $a = 2q$. The Bohr–Sommerfeld area inside the separatrix,

$$\oint p\,\mathrm{d}\varphi = 2\sqrt{2I\lambda}\int_0^{2\pi} \sin\frac{\varphi}{2}\,\mathrm{d}\varphi = 8\sqrt{2I\lambda}, \tag{15}$$

gives the number of librating levels

$$N_{\mathrm{lib}} \simeq \frac{8\sqrt{2I\lambda}}{2\pi\hbar} = \frac{4}{\pi}\sqrt{q}; \tag{16}$$

for $\lambda I/\hbar^2 = 50, 200, 800, 3200$ this is $12.7, 25.5, 50.9, 101.9$, against $13, 25, 51, 102$ counted.

2.4. The entanglement entropy is a Shannon entropy

The reduced density matrix of rotor 1 is $\rho_1 = \mathrm{Tr}_2\,\rho = \sum_m \langle m|_2\, \rho\, |m\rangle_2$, and on a single dyad

$$\mathrm{Tr}_2\, |a,b\rangle\langle c,d| = \delta_{bd}\; |a\rangle\langle c|\,. \tag{17}$$

An eigenstate in sector M is $|\Psi\rangle = \sum_k c_k \left|\tfrac{M}{2} + k, \tfrac{M}{2} - k\right\rangle$. Two different values of k carry different rotor-2 labels, so by (17) every cross term drops out:

$$\rho_1 = \sum_k |c_k|^2 \left|\tfrac{M}{2} + k\right\rangle\!\left\langle \tfrac{M}{2} + k\right|, \qquad \boxed{S_q = -\sum_k p_k \ln p_k, \qquad p_k = |c_k|^2} \tag{18}$$

exactly. The entanglement entropy is the Shannon entropy of the Fourier coefficients of a Mathieu function. Conservation of J fixes the Schmidt basis to be the angular-momentum basis, whatever the energy and the coupling; this fact will be decisive in Secs. 4 and 7.

2.5. Exchange symmetry, doublets and branch states

The exchange $P : \theta_1 \leftrightarrow \theta_2$ maps $k \to -k$ and commutes with H; in the Mathieu language it is $z \to -z$, with ce and se its even and odd eigenfunctions, so every exact eigenstate has $c_{-k} = \pm c_k$. Above the separatrix the levels form doublets of opposite parity, split by an amount that vanishes as $\lambda^{|n|}$ at weak coupling and exponentially in the semiclassical regime: the two members are the even and odd combinations of two *branch* states $|\Psi_{\pm n}\rangle$ describing relative rotation in the two senses, $|\Psi_{-n}\rangle = P\,|\Psi_n\rangle$. On times short compared with the inverse splitting a branch state is effectively stationary.

If the branch weights $q_k = |\,\langle k|\Psi_n\rangle\,|^2$ are supported on $k > 0$ (up to exponentially small tails), the parity eigenstate $(|\Psi_n\rangle \pm |\Psi_{-n}\rangle)/\sqrt{2}$ has $p_{\pm k} = q_k/2$ and, by (18),

$$S_{\text{exact}} = -2\sum_{k>0} \frac{q_k}{2} \ln \frac{q_k}{2} = \ln 2 + S_{\text{branch}}. \tag{19}$$

Note that no cross term can contribute, because ρ_1 is diagonal; only the overlap of the two supports could. Classically the two branches are the two invariant tori $p > 0$ and $p < 0$ of the level set, which the microcanonical distribution weights equally (Sec. 3), so the same $\ln 2$ appears on both sides. Below the separatrix there is one torus and the levels are non-degenerate.

2.6. Parity sectors in the plane-wave basis

Because the doublets are exponentially close, a dense eigensolver applied to (11) returns arbitrary mixtures of their members. We diagonalise in the eigenspaces of P separately. For even M the even sector has basis $\{\chi_0, (\chi_k+\chi_{-k})/\sqrt{2}\}_{k\geq 1}$, in which $\langle\chi_0|\, h\, \frac{1}{\sqrt{2}}(\chi_1+\chi_{-1})\rangle = \lambda/\sqrt{2}$ while all other off-diagonal elements are $\lambda/2$; the odd sector has basis $\{(\chi_k-\chi_{-k})/\sqrt{2}\}_{k\geq 1}$ with off-diagonal elements $\lambda/2$. Every even-M sector is identical to $M = 0$ up to the constant $\hbar^2 M^2/4I$, and every odd-M sector to $M = 1$; all numerics below use $M = 0$ and $\hbar = I = 1$.

3. The classical reduced distribution

3.1. Microcanonical distribution and its marginal

Following [1], the classical counterpart of an eigenstate with definite energy and total angular momentum is the microcanonical distribution on the corresponding level set,

$$W_{\text{cl}}(\theta_1, p_1; \theta_2, p_2) = \mathcal{C}\ \delta(E - H)\ \delta(J - p_1 - p_2)\,, \tag{20}$$

normalised to one. It is invariant under the flow and uniform in the angle variables of each invariant torus. For $E_{\text{rel}} > \lambda$ the level set consists of two tori ($p > 0$ and $p < 0$) of equal weight; for $|E_{\text{rel}}| < \lambda$ of one.

The reduced distribution of rotor 1 is $W_{\text{cl}}(\theta_1, p_1) = \int \mathrm{d}\theta_2 \mathrm{d}p_2\, W_{\text{cl}}$. The second delta function sets $p_2 = J - p_1$, after which, with $p = p_1 - J/2$ and $\varphi = \theta_1 - \theta_2$,

$$W_{\text{cl}}(\theta_1, p_1) = \mathcal{C} \int_0^{2\pi} \mathrm{d}\varphi\ \delta\!\left(E_{\text{rel}} - \frac{p^2}{I} - \lambda\cos\varphi\right). \tag{21}$$

The argument vanishes at $\cos\varphi = \xi \equiv (E_{\text{rel}} - p^2/I)/\lambda$, i.e. at the two angles $\varphi = \pm\arccos\xi$ when $|\xi| < 1$, where the derivative has modulus $\lambda\sqrt{1-\xi^2}$. Hence

$$\boxed{W_{\text{cl}}(\theta_1, p_1) = \frac{2\mathcal{C}}{\sqrt{\lambda^2 - (E_{\text{rel}} - p^2/I)^2}}\ \Theta\big(\lambda - \big|E_{\text{rel}} - p^2/I\big|\big)} \tag{22}$$

independent of θ_1. The two roots $\varphi = \pm\arccos\xi$ — two relative angles compatible with one momentum of rotor 1 — are the two classical branches whose quantum interference is the subject of Secs. 4 and 7.

3.2. The support and its cusp at the separatrix

The band $|E_{\rm rel} - p^2/I| \le \lambda$ is $|p| \le \sqrt{I(E_{\rm rel}+\lambda)}$ in libration and $\sqrt{I(E_{\rm rel}-\lambda)} \le |p| \le \sqrt{I(E_{\rm rel}+\lambda)}$ in rotation, of total length

$$L(E_{\rm rel}) = \begin{cases} 2\sqrt{I(E_{\rm rel}+\lambda)}, & E_{\rm rel} < \lambda, \\ 2\left[\sqrt{I(E_{\rm rel}+\lambda)} - \sqrt{I(E_{\rm rel}-\lambda)}\right], & E_{\rm rel} > \lambda. \end{cases} \tag{23}$$

L increases through libration, reaches its maximum $2\sqrt{2I\lambda}$ at the separatrix and decreases above it with $\mathrm{d}L/\mathrm{d}E_{\rm rel} \to -\infty$ as $E_{\rm rel} \to \lambda^+$: a cusp maximum. Since the entropy of a distribution on a set of length L is at most $\ln(L/\hbar)$, this is the origin of the entropy maximum of Sec. 5.

3.3. Normalisation and the classical period

The momentum density is $w(p_1) = 2\pi W_{\rm cl}$. Its normalisation needs the band integral $B(E_{\rm rel}) = \int_{\text{band},\,p>0} \mathrm{d}p\,[\lambda^2 - (E_{\rm rel} - p^2/I)^2]^{-1/2}$. With $E_{\rm rel} - p^2/I = \lambda\cos u$, $B = \frac{\sqrt{I}}{2}\int \mathrm{d}u\,(E_{\rm rel} - \lambda\cos u)^{-1/2}$, and the standard substitutions give (Appendix A)

$$B = \begin{cases} \sqrt{\dfrac{I}{E_{\rm rel}+\lambda}}\,\mathsf{K}(k), \quad k^2 = \dfrac{2\lambda}{E_{\rm rel}+\lambda}, & E_{\rm rel} > \lambda, \\ \sqrt{\dfrac{I}{2\lambda}}\,\mathsf{K}(\kappa), \quad \kappa^2 = \dfrac{E_{\rm rel}+\lambda}{2\lambda}, & |E_{\rm rel}| < \lambda, \end{cases} \tag{24}$$

with K the complete elliptic integral of the first kind. In either regime the total weight is $2B$, so

$$\mathcal{C} = \frac{1}{8\pi B}, \qquad w(p_1) = \frac{1}{2B}\,\frac{1}{\sqrt{\lambda^2 - (E_{\rm rel} - p^2/I)^2}}. \tag{25}$$

With $\dot\varphi = 2p/I$ the same substitution shows that the period of the relative motion is $T = 2B$: $w(p_1)\,\mathrm{d}p_1$ is the fraction of the period spent in $\mathrm{d}p_1$. At the separatrix K diverges logarithmically.

3.4. Boltzmann entropy and the elementary cell

The Boltzmann entropy of the reduced distribution, $S_{\rm cl} = -\int \mathrm{d}\theta_1 \mathrm{d}p_1\, W_{\rm cl}\ln(W_{\rm cl}\Delta)$, requires a cell Δ. On a compact configuration space Δ is not free: $p_1 = \hbar m_1$, and each quantum state occupies the strip $2\pi \times \hbar = h$. With $\Delta = h$ and $W_{\rm cl} = w/2\pi$,

$$S_{\rm cl} = -\int \mathrm{d}p_1\; w \ln\left(w\hbar\right), \tag{26}$$

a differential entropy in units of $\hbar$. The object with the same status as (18) is the Shannon entropy of the cell weights,

$$w_m = \int_{\hbar(m-1/2)}^{\hbar(m+1/2)} w(p_1)\,\mathrm{d}p_1, \qquad S_{\rm cl}^{\rm bin} = -\sum_m w_m \ln w_m. \tag{27}$$

The comparison between (27) and (18) contains no adjustable constant. Writing $\varrho = w/w_m$ on cell m,

$$S_{\rm cl} = S_{\rm cl}^{\rm bin} - \sum_m w_m\, D_{\rm KL}\big(\varrho \,\|\, \hbar^{-1}\big) \le S_{\rm cl}^{\rm bin}, \tag{28}$$

with $D_{\rm KL}$ the relative entropy with respect to the uniform density on the cell. The difference comes from the cells that contain the inverse-square-root singularities at the band edges and decreases as (band width in cells)$^{-1/2}$.

4. Deep rotation: an exactly solvable limit and the constant $1 - \ln 2$

4.1. Reduction to a Wannier–Stark ladder

Consider a level with large relative angular momentum, $k_0 = n/2 \gg 1$, and write $k = k_0 + j$. Then

$$\frac{\hbar^2}{I}k^2 = \frac{\hbar^2}{I}k_0^2 + F\,j + \frac{\hbar^2}{I}j^2, \qquad F \equiv \frac{2\hbar^2 k_0}{I} = \frac{\hbar^2 n}{I}. \tag{29}$$

For $|j| \lesssim \gamma$ (defined below) the quadratic term is smaller than the linear one by $|j|/n \le \gamma/n$. Dropping it, (11) becomes, up to a constant,

$$h_{\rm WS}\, c_j = F\, j\, c_j + \frac{\lambda}{2}\,(c_{j+1} + c_{j-1})\,, \tag{30}$$

a particle hopping on a lattice in a uniform field: the Wannier–Stark problem [13, 14]. Try $c_j = J_{j-s}(x)$ with $s \in \mathbb{Z}$. The recurrence $J_{\nu-1}(x) + J_{\nu+1}(x) = (2\nu/x)J_\nu(x)$ gives

$$\frac{\lambda}{2}\,(c_{j+1} + c_{j-1}) = \frac{\lambda(j-s)}{x}c_j, \qquad h_{\rm WS}c_j = \left[Fj + \frac{\lambda(j-s)}{x}\right] c_j, \tag{31}$$

which is independent of j if and only if $x = -\lambda/F$; the eigenvalue is then Fs. Using $J_\nu(-x) = (-1)^\nu J_\nu(x)$,

$$|c_j|^2 = J_{j-s}^2(\gamma), \qquad \boxed{\gamma \equiv \frac{\lambda}{F} = \frac{\lambda I}{\hbar^2 n} = \frac{2\alpha}{n}} \tag{32}$$

normalised by the Neumann identity $\sum_j J_j^2(\gamma) = 1$. The eigenvectors are translates of one another, so all have the same entropy. The neglected term is small when $\gamma/n = \lambda/4E_{\rm rel} \ll 1$, deep in the rotation regime. Exact diagonalisation confirms (32) pointwise to 3×10^{-3} at $\gamma = 2, 5$ ($\gamma/n = 0.01$) and the entropy to four or five digits for $n = 100$–400 (Fig. 1a).

The Bessel eigenvector is a branch state: it is centred on $k_0 > 0$ and its weight on $k < 0$ is negligible, so the parity eigenstate has entropy $\ln 2 + S_{\rm branch}$ by (19).

4.2. The branch entropy as a function of one variable

From (18) and (32) the branch entropy in deep rotation is a universal function of γ alone,

$$\boxed{S_{\rm branch}(\gamma) = -\sum_{j\in\mathbb{Z}} J_j^2(\gamma)\ln J_j^2(\gamma)} \tag{33}$$

Small γ. With $J_0^2 = 1 - \gamma^2/2 + \dots$ and $J_{\pm1}^2 = \gamma^2/4 + \dots$,

$$S_{\rm branch} \simeq 2\cdot\frac{\gamma^2}{4}\ln\frac{4}{\gamma^2} + \frac{\gamma^2}{2} = \gamma^2\left[\ln\frac{2}{\gamma} + \frac{1}{2}\right]. \tag{34}$$

This is second-order perturbation theory in λ: $\gamma^2/4$ is the weight transferred to each neighbour $n \pm 2$, and the term $\gamma^2/2$ comes from the depletion $1 - \gamma^2/2$ of the unperturbed component, i.e. from the $O(\lambda^2)$ normalisation of the perturbed state, which must be kept in the density matrix [5].

Large γ. The weights become many and the classical description applies; we treat this next.

4.3. The classical counterpart: the arcsine law

In the same limit, linearising p^2/I about $p_0 = \sqrt{IE_{\rm rel}} = \hbar n/2$ in (22) gives $w \propto [a^2 - (p - p_0)^2]^{-1/2}$ with half-width $a = I\lambda/2p_0$, i.e. $a/\hbar = \gamma$. The classical weights are the arcsine law,

$$w(j) = \frac{1}{\pi\sqrt{\gamma^2 - j^2}}, \qquad |j| < \gamma, \tag{35}$$

whose differential entropy is (Appendix A)

$$S_{\rm cl}^{\rm branch} = \ln\frac{\pi\gamma}{2}, \qquad S_{\rm cl}^{\rm two\ bands} = \ln(\pi\gamma) = \ln\left[\frac{\pi\lambda}{2\hbar}\sqrt{\frac{I}{E_{\rm rel}}}\right]. \tag{36}$$

This is the rotor counterpart of (2): the entropy grows logarithmically with the coupling and decreases logarithmically with the relative energy.

4.4. Interference fringes in the quantum weights

The quantum weights oscillate about the arcsine law. The Debye expansion of the Bessel function [15], with $j = \gamma\cos\beta$, $0 < \beta < \pi$, reads $J_j(\gamma) \simeq \sqrt{2/(\pi\gamma\sin\beta)}\,\cos(\gamma\sin\beta - j\beta - \pi/4)$. Squaring,

$$J_j^2(\gamma) \simeq w(j)\,[1 + \cos\Phi_j]\,, \qquad \Phi_j = 2\sqrt{\gamma^2 - j^2} - 2j\arccos\frac{j}{\gamma} - \frac{\pi}{2}, \tag{37}$$

fringes of *full* contrast about the classical weights. Their local wavenumber is

$$\frac{{\rm d}\Phi_j}{{\rm d}j} = -\frac{2j}{\sqrt{\gamma^2 - j^2}} - 2\arccos\frac{j}{\gamma} + \frac{2j}{\sqrt{\gamma^2 - j^2}} = -2\beta, \tag{38}$$

ranging over $(-2\pi, 0)$ across the band.

The same structure holds at any energy. In the k representation the eigenvalue equation is the difference equation $\frac{\hbar^2}{I}k^2 c_k + \frac{\lambda}{2}(c_{k+1} + c_{k-1}) = E_{\rm rel}c_k$. The WKB ansatz $c_k = A_k e^{i\chi_k}$ with slowly varying A_k and local wavenumber $\phi_k = \chi_{k+1} - \chi_k$ gives at leading order $E_{\rm rel} - \hbar^2k^2/I = \lambda\cos\phi_k$: the local wavenumber is the classical relative angle at momentum $\hbar k$, and the two roots $\pm\phi_k$ are the two branches of (22). The Casoratian $\frac{\lambda}{2}(c_k\bar{c}_{k+1} - c_{k+1}\bar{c}_k)$ of two solutions is independent of k, which for one WKB wave gives $|A_k|^2\lambda\sin\phi_k = {\rm const}$, i.e. $|A_k|^2 \propto [\lambda^2 - (E_{\rm rel} - \hbar^2k^2/I)^2]^{-1/2}$: the classical density. A real eigenvector superposes the two waves with equal amplitude, so

$$p_k \simeq w_k\,(1 + \cos\Phi_k)\,, \qquad \Phi_{k+1} - \Phi_k = 2\phi_k. \tag{39}$$

The classical distribution is the fringe average of the quantum one.

4.5. The entropy deficit

Insert (39) into (18):

$$S_q = S_{\rm cl}^{\rm bin} - \sum_k w_k \cos\Phi_k \ln w_k - \sum_k w_k(1+\cos\Phi_k)\ln(1+\cos\Phi_k). \tag{40}$$

In the second term $\cos\Phi_k$ oscillates on the scale of a few cells while $w_k \ln w_k$ varies on the scale of the band, so it vanishes as the band widens. In the third the phases $\Phi_k \bmod 2\pi$ are equidistributed wherever $2\phi_k$ is not close to a multiple of 2π, and the sum becomes a phase average (Appendix A):

$$S_{\rm cl} - S_q \;\longrightarrow\; \frac{1}{2\pi}\int_0^{2\pi}(1+\cos\Phi)\ln(1+\cos\Phi)\,{\rm d}\Phi = 1-\ln 2, \tag{41}$$

that is,

$$\boxed{S_q = S_{\rm cl} - (1-\ln 2) + o(1)} \qquad 1-\ln 2 = 0.30685\ldots \tag{42}$$

The limiting deficit is universal: it depends only on the full contrast of the two-branch interference.

Rate of approach. Equidistribution fails near the band edges, where $\phi_k \to 0$ or π and the fringes become stationary (the caustics of the momentum-space WKB). Near $j = \gamma - \delta$, $\beta \simeq \sqrt{2\delta/\gamma}$ and the phase accumulated over the last δ cells is $\int 2\beta\,{\rm d}j \simeq \frac{4\sqrt{2}}{3}\delta^{3/2}/\sqrt{\gamma}$, which reaches 2π at $\delta \sim \gamma^{1/3}$; the arcsine weight within that distance of the edge is $\simeq \pi^{-1}\sqrt{2\delta/\gamma} \sim \gamma^{-1/3}$. Hence $S_{\rm cl} - S_q = 1 - \ln 2 - O(\gamma^{-1/3})$.

4.6. Numerical confirmation

Because (32) is explicit, (42) can be tested to very large γ. Figure 1(b) shows $S_{\rm cl}^{\rm bin} - S_q$ for $\gamma = 10$ to 10^6: 0.233 at $\gamma = 10^3$, 0.270 at 10^4, 0.288 at 10^5 and 0.298 at 10^6; the distance to $1 - \ln 2$ falls by a factor close to $2.15 = 10^{1/3}$ per decade.

4.7. Interpretation

In the oscillator problem no conservation law singles out a Schmidt basis: the reduced density matrix of [1] is doubly degenerate, its eigenvalues follow the smooth arcsine law, and the two-branch interference is absorbed into the eigenvectors. For the rotors, conservation of J forces the Schmidt basis to be the angular-momentum basis (Sec. 2.4), so the interference cannot be absorbed into the eigenvectors and appears in the eigenvalues, as the fringes (39). The leading logarithms agree; the constant terms differ by $1 - \ln 2$. Equivalently, the classical formula (26) reproduces the quantum entropy if the cell is taken to be $eh/2$ rather than h, since $\ln(e/2) = 1 - \ln 2$. Section 7 locates the same interference in phase space.

5. Generic energies and the separatrix

Away from deep rotation we compare (18), computed by exact diagonalisation in the parity sectors, with (27), computed by adaptive quadrature of (22) over each cell. Figure 2 shows both for $\lambda I/\hbar^2 = 50, 200, 800$ as functions of $E_{\rm rel}/\lambda$.

Agreement and deficit. The curves track each other over the whole range. The quantum entropy lies below the classical one throughout libration, by about 0.2, except within a few levels

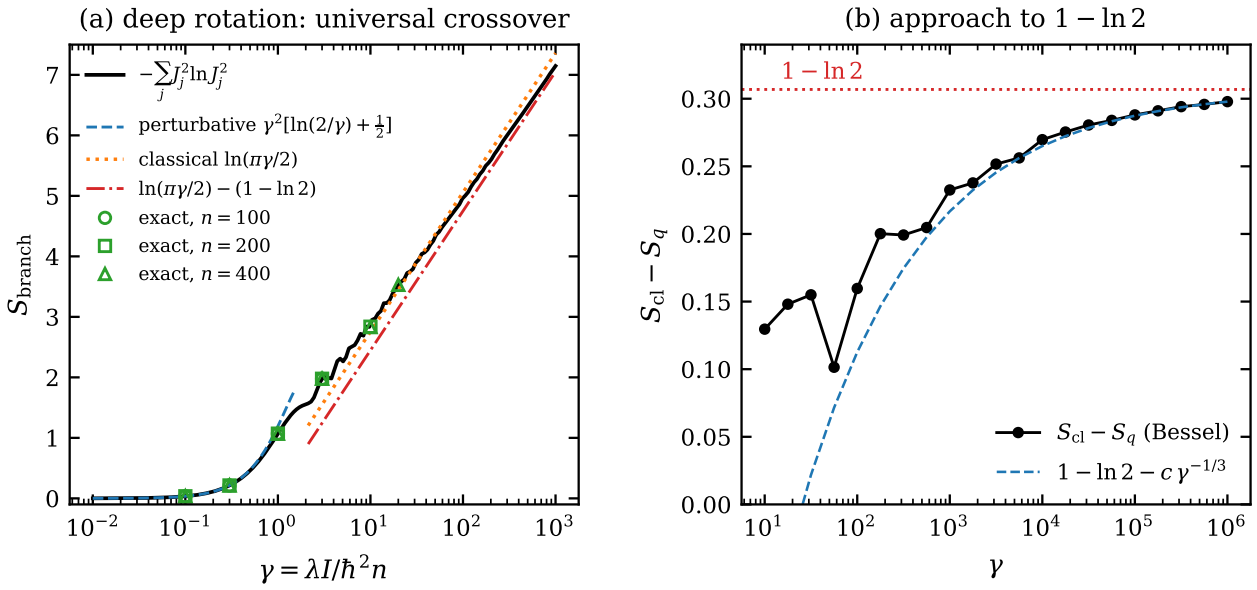


Figure 1: Deep rotation. (a) The branch entropy as a universal function of $\gamma = \lambda I/\hbar^2 n$: the Bessel result (33) (black), its small-γ (second-order perturbative) limit (34) (blue dashed), the classical arcsine entropy (orange dotted) and the classical entropy minus $1-\ln 2$ (red dash-dotted). Open symbols: exact diagonalisation at $n = 100, 200, 400$. (b) The deficit $S_{\rm cl}^{\rm bin} - S_q$ from the Bessel weights, approaching $1-\ln 2$ as $\gamma^{-1/3}$.

of the bottom of the well, where the band covers only a handful of cells; above the separatrix it lies below by a smaller amount that grows with λ. Relative to $S \approx 3$–4 this is a few percent. Figure 3(a) shows the mechanism: at $\lambda I/\hbar^2 = 800$, $E_{\rm rel}/\lambda = 1.54$, the quantum weights oscillate about the classical envelope with full contrast, the total-variation distance between the two distributions is 0.33, yet the entropies differ by less than 1%. Figure 3(b) shows the deficit averaged over the energy windows $0.2 \le E_{\rm rel}/\lambda \le 0.7$ (libration) and $1.5 \le E_{\rm rel}/\lambda \le 2.5$ (rotation) as $\lambda I/\hbar^2$ grows from 200 to 51 200: in libration it rises from 0.18 to 0.24, in rotation from 0.01 to 0.15. Both increase monotonically towards $1-\ln 2$, more slowly than in the pure Wannier–Stark case, because the bands contain only 10–500 cells and the caustic regions carry a large share of the weight.

The separatrix. Both entropies are maximal near $E_{\rm rel} = \lambda$; for $\lambda I/\hbar^2 = 50$, 200 and 800 the quantum maximum is at $E_{\rm rel}/\lambda = 1.13$, 1.12 and 0.90. The classical reason is the cusp of the support (23). On the quantum side this is where the non-degenerate librational levels give way to the rotational doublets. The deviation in Fig. 2(b) changes abruptly there, because the fringe structure changes: below the separatrix the two interfering branches are the two halves of one libration orbit, above it the two halves of one rotation orbit, and the caustics move from $p = \pm\sqrt{I(E_{\rm rel}+\lambda)}$ alone to include $p = \pm\sqrt{I(E_{\rm rel}-\lambda)}$. The separatrix is the classical locus of an excited-state quantum phase transition [16, 17, 18], and the entropy maximum should be understood in that context.

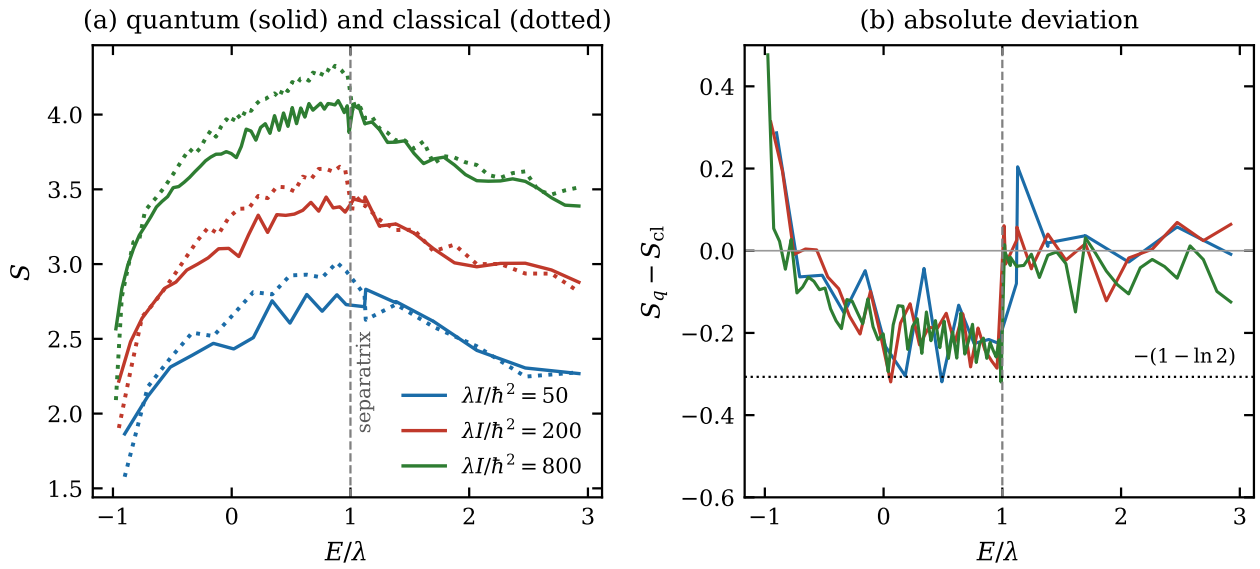


Figure 2: (a) Quantum entanglement entropy (solid) and classical coarse-grained entropy (dotted) as functions of $E_{\rm rel}/\lambda$ for $\lambda I/\hbar^2 = 50$, 200 and 800. The dashed line marks the separatrix. (b) The deviation $S_q - S_{\rm cl}^{\rm bin}$, with the asymptotic value $-(1-\ln 2)$ of (42) shown dotted.

6. Deep libration

6.1. Harmonic approximation

For $\lambda I/\hbar^2 \gg 1$ the ground state of h is localised near $\varphi = \pi$. With $x = \varphi - \pi$, $\lambda\cos\varphi = -\lambda + \frac{1}{2}\lambda x^2 - \frac{1}{24}\lambda x^4 + \dots$,

$$h = -\lambda + \frac{p^2}{2\mu} + \frac{1}{2}\mu\omega^2 x^2 - g x^4 + \dots, \qquad \mu = \frac{I}{2}, \quad \mu\omega^2 = \lambda, \quad g = \frac{\lambda}{24}, \tag{43}$$

so $\omega = \sqrt{2\lambda/I}$. Neglecting g, the ground state is $u_0 \propto \exp(-\mu\omega x^2/2\hbar)$, with $\langle p^2\rangle = \mu\hbar\omega/2$. Its Fourier coefficients are, up to the phase $e^{-ik\pi}$, a Gaussian in $k = p/\hbar$ with variance

$$\sigma_0^2 = \frac{\langle p^2\rangle}{\hbar^2} = \frac{\mu\omega}{2\hbar} = \frac{1}{4}\sqrt{\frac{2\lambda I}{\hbar^2}}. \tag{44}$$

6.2. Entropy of a sampled Gaussian

For $p_k = e^{-k^2/2\sigma^2}/Z$ on the integers, Poisson summation gives

$$Z = \sqrt{2\pi}\,\sigma\left[1 + 2\sum_{r\geq 1} e^{-2\pi^2\sigma^2 r^2}\right], \qquad \sum_k k^2 p_k = \sigma^2\left[1 + O\big(\sigma^2 e^{-2\pi^2\sigma^2}\big)\right]. \tag{45}$$

Since $-\ln p_k = \ln Z + k^2/2\sigma^2$,

$$S = \frac{1}{2}\ln\left(2\pi e\sigma^2\right) + O\Big(\sigma^2 e^{-2\pi^2\sigma^2}\Big). \tag{46}$$

The discreteness correction is exponentially small. With $\sigma = \sigma_0$,

$$S_0 = \frac{1}{2}\ln\left(2\pi e\cdot\frac{1}{4}\sqrt{\frac{2\lambda I}{\hbar^2}}\right) = \frac{1}{4}\ln\frac{\lambda I}{\hbar^2} + \frac{1}{2}\ln\frac{\pi e}{\sqrt{2}}. \tag{47}$$

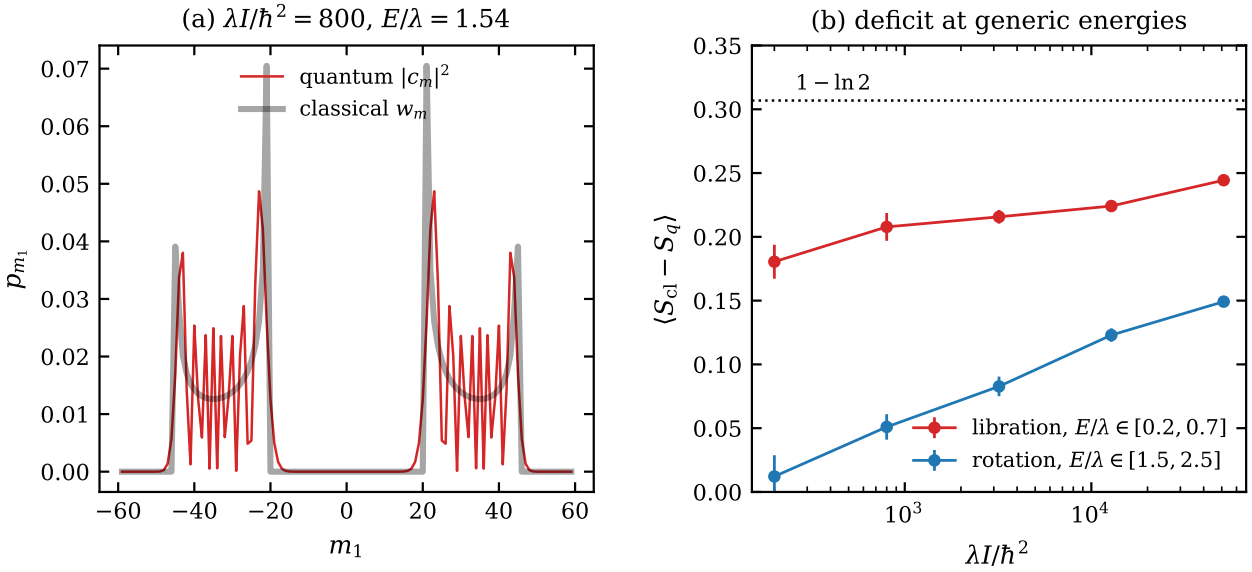


Figure 3: (a) Momentum distribution of rotor 1 at $\lambda I/\hbar^2 = 800$, $E_{\rm rel}/\lambda = 1.54$: the quantum weights (red) oscillate with full contrast about the classical cell weights (grey), as (39) predicts. (b) Deficit $\langle S_{\rm cl}^{\rm bin} - S_q \rangle$ averaged over energy windows in libration and in rotation, against λ; error bars are standard errors over the window.

6.3. The leading anharmonic correction

The quartic term narrows the momentum distribution. The virial theorem, $\langle [h, xp] \rangle = 0$, gives $2\langle T \rangle = \langle xV' \rangle$. With $V = V_2 - V_4$, $V_2 = \frac{1}{2}\mu\omega^2 x^2$, $V_4 = gx^4$, one has $xV' = 2V_2 - 4V_4$, so $\langle V_2 \rangle = \langle T \rangle + 2\langle V_4 \rangle$ and $E = \langle T \rangle + \langle V_2 \rangle - \langle V_4 \rangle$ gives $\langle T \rangle = \frac{1}{2}(E - \langle V_4 \rangle)$. To first order in g, $E = \frac{1}{2}\hbar\omega - g\langle x^4 \rangle_0$ and $\langle V_4 \rangle = g\langle x^4 \rangle_0$, with $\langle x^4 \rangle_0 = 3\hbar^2/4\mu^2\omega^2$. Therefore

$$\langle p^2 \rangle = 2\mu\langle T \rangle = \frac{\mu\hbar\omega}{2} - \frac{3g\hbar^2}{2\mu\omega^2} = \frac{\mu\hbar\omega}{2} - \frac{\hbar^2}{16}, \qquad \sigma^2 = \sigma_0^2 - \frac{1}{16}. \tag{48}$$

The quartic term also makes the distribution non-Gaussian, but this does not change the entropy at first order: if $p = \phi_\sigma(1 + \epsilon\eta)$ with ϕ_σ the Gaussian of the exact variance, then η is orthogonal (with weight ϕ_σ) to 1, k and k^2, and $-\sum p \ln p = S[\phi_\sigma] - \epsilon \sum \phi_\sigma \eta(1 + \ln \phi_\sigma) + O(\epsilon^2) = S[\phi_\sigma] + O(\epsilon^2)$, because $\ln \phi_\sigma$ is a quadratic polynomial in k. Expanding $\frac{1}{2}\ln(\sigma_0^2 - \frac{1}{16})$,

$$\boxed{S_0 = \frac{1}{4}\ln\frac{\lambda I}{\hbar^2} + \frac{1}{2}\ln\frac{\pi e}{\sqrt{2}} - \frac{1}{8\sqrt{2\lambda I/\hbar^2}} + O\left(\frac{\hbar^2}{\lambda I}\right)} \tag{49}$$

with $\frac{1}{2}\ln(\pi e/\sqrt{2}) = 0.89908$. For $\lambda I/\hbar^2 = 10$, 40, 160, 640, 2560 and 10240 the residual after the harmonic terms is -0.03027, -0.01451, -0.00712, -0.00353, -0.00176 and -0.00088, and its ratio to the predicted correction is 1.083, 1.039, 1.019, 1.009, 1.005 and 1.002 (Fig. 4).

7. Wigner and Husimi distributions

We now ask what happens if the classical microcanonical distribution (20) is replaced by a quantum phase-space distribution of the eigenstate itself. Throughout this section ℓ denotes momentum in units of $\hbar$.

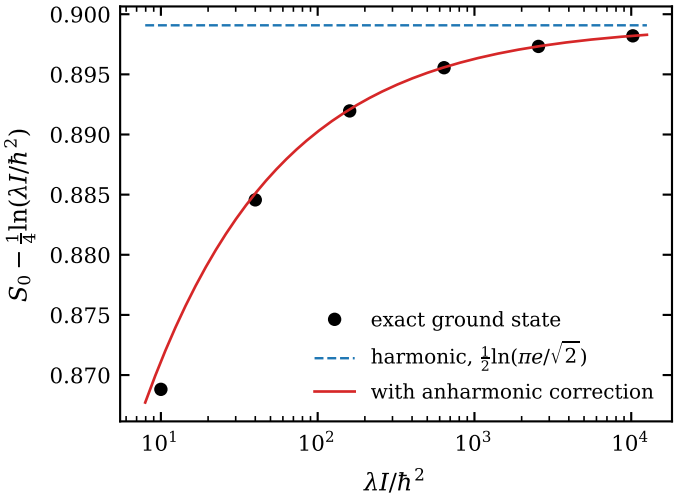


Figure 4: Deep libration: $S_0 - \frac{1}{4}\ln(\lambda I/\hbar^2)$ for the ground state from exact diagonalisation (points), the harmonic constant $\frac{1}{2}\ln(\pi e/\sqrt{2})$ (dashed), and (49) with the anharmonic correction (solid).

7.1. The Wigner function on a circle

For a single rotor with density matrix ρ and $\langle\theta|m\rangle = e^{im\theta}/\sqrt{2\pi}$, the Wigner function is [19, 20, 21]

$$W_\rho(\theta,\ell) = \frac{1}{4\pi}\int_{-2\pi}^{2\pi} \mathrm{d}\xi\; e^{-i\ell\xi}\,\left\langle \theta+\tfrac{\xi}{2}\right|\rho\left|\theta-\tfrac{\xi}{2}\right\rangle, \qquad \ell\in\tfrac{1}{2}\mathbb{Z}. \tag{50}$$

The kernel is 4π-periodic in ξ (a shift of ξ by 4π moves $\theta\pm\xi/2$ by $\pm 2\pi$), so its Fourier variable ℓ runs over the half-integers. Inserting the basis,

$$\left\langle \theta+\tfrac{\xi}{2}\right|\rho\left|\theta-\tfrac{\xi}{2}\right\rangle = \frac{1}{2\pi}\sum_{m,m'}\rho_{mm'}\,e^{i(m-m')\theta}\,e^{i(m+m')\xi/2}, \tag{51}$$

and since $\int_{-2\pi}^{2\pi} e^{is\xi}\mathrm{d}\xi = 4\pi\delta_{s0}$ for $s\in\frac{1}{2}\mathbb{Z}$,

$$W_\rho(\theta,\ell) = \frac{1}{2\pi}\sum_{m+m'=2\ell}\rho_{mm'}\,e^{i(m-m')\theta}. \tag{52}$$

From (52): (i) W_ρ is real for Hermitian ρ; (ii) $\sum_{\ell\in\frac{1}{2}\mathbb{Z}} W_\rho(\theta,\ell) = \langle\theta|\rho|\theta\rangle$; (iii) $\int_0^{2\pi}\mathrm{d}\theta\, W_\rho(\theta,\ell) = \rho_{\ell\ell}$ for $\ell\in\mathbb{Z}$ and 0 for $\ell\in\mathbb{Z}+\frac{1}{2}$; (iv) $\mathrm{Tr}(\rho\sigma) = 2\pi\sum_\ell\int\mathrm{d}\theta\, W_\rho W_\sigma$, since the θ integral forces $m-m' = n'-n$, which with $m+m' = n+n'$ gives $m=n'$, $m'=n$. The half-integer rows have zero momentum marginal: they carry only coherences.

For two rotors the same construction in each variable gives

$$W(\theta_1,\ell_1;\theta_2,\ell_2) = \frac{1}{(2\pi)^2}\sum_{\substack{m_1+m_1'=2\ell_1\\ m_2+m_2'=2\ell_2}}\rho_{m_1m_2;m_1'm_2'}\,e^{i(m_1-m_1')\theta_1+i(m_2-m_2')\theta_2}. \tag{53}$$

Integrating over θ_2 sets $m_2' = m_2$, after which the sum over ℓ_2 sums over m_2 once:

$$\sum_{\ell_2\in\frac{1}{2}\mathbb{Z}}\int_0^{2\pi}\mathrm{d}\theta_2\; W(\theta_1,\ell_1;\theta_2,\ell_2) = \frac{1}{2\pi}\sum_{m_1+m_1'=2\ell_1}(\rho_1)_{m_1m_1'}e^{i(m_1-m_1')\theta_1} = W_{\rho_1}(\theta_1,\ell_1). \tag{54}$$

Integrating the Wigner function over the second rotor is the partial trace.

7.2. Eigenstates: the Wigner analogue of the DKS construction

For an eigenstate $|\Psi\rangle = \sum_k c_k \left|\frac{M}{2}+k, \frac{M}{2}-k\right\rangle$ the matrix elements are $\rho_{m_1 m_2; m_1' m_2'} = c_k \bar{c}_{k'}$ with $m_1 = \frac{M}{2}+k$, $m_2 = \frac{M}{2}-k$, and similarly for the primes. The constraints in (53) become $2\ell_1 = M + k + k'$, $2\ell_2 = M - k - k'$, and the phase is $(k-k')(\theta_1 - \theta_2)$. Hence

$$W(\theta_1, \ell_1; \theta_2, \ell_2) = \frac{1}{2\pi}\, \delta_{\ell_1+\ell_2, M}\; W_{\mathrm{rel}}\big(\theta_1 - \theta_2,\; \ell_1 - \tfrac{M}{2}\big)\,, \qquad W_{\mathrm{rel}}(\varphi, \ell) = \frac{1}{2\pi} \sum_{k+k'=2\ell} c_k \bar{c}_{k'}\, e^{i(k-k')\varphi}, \tag{55}$$

where W_{rel} is the Wigner function (52) of the relative wave function $u(\varphi)$. The classical distribution (20) has the same skeleton: in the variables (8),

$$W_{\mathrm{cl}} = \frac{1}{2\pi}\, \delta(J - p_1 - p_2)\; \frac{\delta(E_{\mathrm{rel}} - H_{\mathrm{rel}}(\varphi, p))}{4B}, \tag{56}$$

as follows from $\mathcal{C} = 1/8\pi B$ and $\int \mathrm{d}\varphi\, \mathrm{d}p\; \delta(E_{\mathrm{rel}} - H_{\mathrm{rel}}) = 4B$. The factor $1/2\pi$ (uniformity in the centre-of-mass angle) and the conservation of total angular momentum appear identically; the only difference is the relative factor, W_{rel} against the microcanonical density on the pendulum level set.

Now carry out the DKS construction with W in place of W_{cl}. By (54) and (18) the reduced Wigner function is

$$W_1(\theta_1, \ell_1) = \begin{cases} p_k/2\pi, & \ell_1 = \frac{M}{2} + k \in \mathbb{Z} + \frac{M}{2}, \\ 0, & \text{otherwise}, \end{cases} \tag{57}$$

independent of θ_1 and non-negative. Its Boltzmann entropy with the cell fixed in Sec. 3.4 ($2\pi W_1$ is the probability per cell) is

$$-\sum_{\ell_1} \int_0^{2\pi} \mathrm{d}\theta_1\, W_1 \ln\left(2\pi W_1\right) = -\sum_k p_k \ln p_k = S_q. \tag{58}$$

The Wigner version of the classical calculation returns the von Neumann entropy identically, for every eigenstate, at every coupling and every $\hbar$. It contains no semiclassical approximation and no new information. Two remarks. First, the identity requires $W_1 \geq 0$, which in general fails for a reduced Wigner function; here it holds because ρ_1 is diagonal, i.e. because of J conservation. Second, (58) answers the question posed in the Introduction: replacing the classical distribution by the Wigner function does not give a "classical" entropy, it gives the quantum one. The informative object is the relation between W_{rel} and the classical level set, to which we turn.

7.3. Where the classical distribution sits inside the Wigner function

Figure 5 shows W_{rel} for a librating and a rotating level at $\lambda I/\hbar^2 = 200$. Its structure is that described by Berry for one-dimensional eigenstates [22, 23]. The function is large on the classical curve $H_{\mathrm{rel}} = E_{\mathrm{rel}}$ (dashed), with an Airy profile across it, and it oscillates inside: at a point (φ, ℓ) there is a contribution from every chord of the classical curve centred on that point, with a phase equal to the symplectic area between chord and curve in units of $\hbar$. The oscillations make W_{rel} strongly negative: the negativity volume [25] $\sum_\ell \int \mathrm{d}\varphi\, |W_{\mathrm{rel}}| - 1$ is 5.86

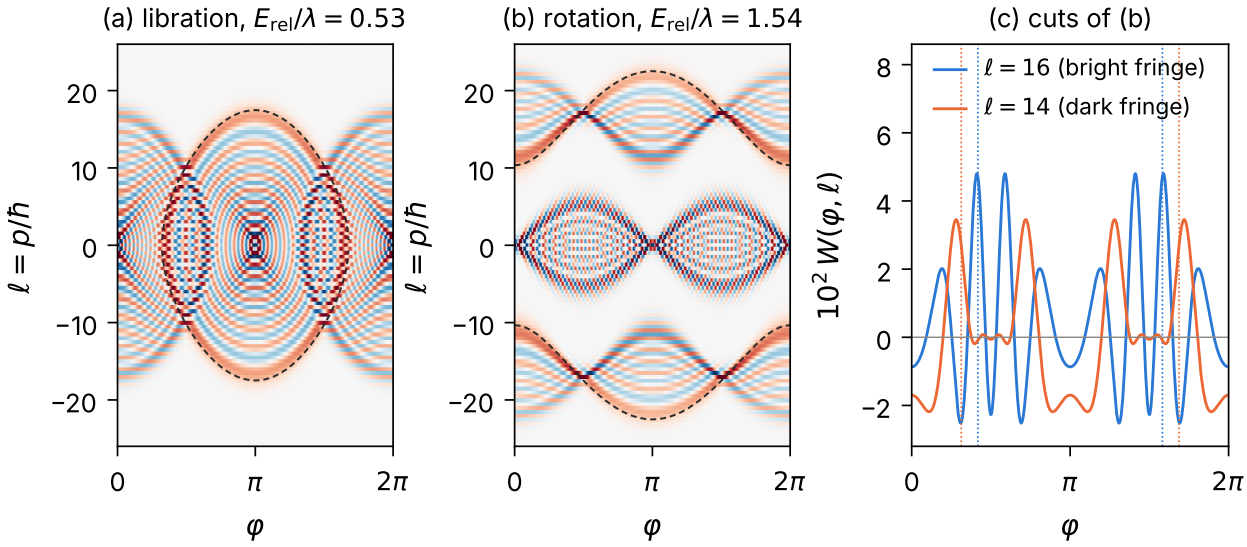


Figure 5: Relative Wigner function $W_{\rm rel}(\varphi, \ell)$, Eq. (55), at $\lambda I/\hbar^2 = 200$ (red positive, blue negative; both integer and half-integer rows are shown). (a) Librating level, $E_{\rm rel}/\lambda = 0.53$; (b) rotating level, $E_{\rm rel}/\lambda = 1.54$ (odd parity: the band near $\ell = 0$ is the interference between the two counter-rotating tori). Dashed: the classical curve $\ell^2 + \lambda\cos\varphi = E_{\rm rel}$. (c) Cuts of (b) at $\ell = 16$ (bright fringe of p_ℓ) and $\ell = 14$ (dark fringe); dotted lines mark the classical angles $\pm\phi_\ell$. Away from them the cut oscillates: these are the chord terms that survive the integration over φ.

and 5.43 for the two states shown. Averaged over a phase-space region large on the scale $\hbar$ but small classically, the chord terms cancel and $W_{\rm rel}$ reduces to the microcanonical density on the curve [22, 24].

The two operations "smooth" and "integrate over φ" do not commute, and this is the phase-space content of Sec. 4. Integrating $W_{\rm rel}$ over φ at fixed integer ℓ collects (a) the two points of the classical curve at momentum ℓ, $\varphi = \pm\phi_\ell$, which give the classical weight w_ℓ, and (b) the horizontal chord joining them, which is centred on $\varphi = 0$ and on $\varphi = \pi$ and gives $w_\ell \cos\Phi_\ell$. Together they produce (39). At $\ell = 14$ in Fig. 5(c), for example, integrating over windows of half-width 0.35 around $\pm\phi_\ell$ gives $+0.022$, around 0 and π gives -0.027, and the full integral is $p_{14} = 7 \times 10^{-4}$, a dark fringe. On half-integer rows the chord contributions at $\varphi = 0$ and π have opposite signs (∓ 0.012 at $\ell = 16.5$) and the marginal vanishes, as property (iii) requires. Smoothing first removes the chord terms and gives the classical distribution; integrating first keeps them. *The constant* $1 - \ln 2$ *is the entropy carried by the chord terms of the Wigner function.*

7.4. The Husimi function and the Wehrl entropy

The natural way to remove the chord terms in a quantum language is the Husimi function [7], the expectation value in coherent states. On the circle we use [26, 27]

$$|\theta, \ell\rangle = \sum_{m\in\mathbb{Z}} \sqrt{g_\sigma(m-\ell)}\; e^{-im\theta}\, |m\rangle\,, \qquad g_\sigma(x) = \frac{e^{-x^2/2\sigma^2}}{\sqrt{2\pi}\,\sigma}, \qquad \theta \in [0, 2\pi),\ \ell \in \mathbb{R}. \tag{59}$$

In the angle representation this is a wave packet centred on θ with mean momentum ℓ, momentum width σ and angular width $1/2\sigma$. The resolution of the identity, $\int_0^{2\pi} \frac{\mathrm{d}\theta}{2\pi} \int_{-\infty}^{\infty} \mathrm{d}\ell\; |\theta, \ell\rangle\langle\theta, \ell| =$

$\sum_m |m\rangle\langle m| \int \mathrm{d}\ell\, g_\sigma(m-\ell) = \mathbb{1}$, holds exactly; the measure $\mathrm{d}\theta\,\mathrm{d}\ell/2\pi$ is the phase space measure with cell h. By Poisson summation the norm is $\langle\theta,\ell|\theta,\ell\rangle = \sum_m g_\sigma(m-\ell) = 1 + 2\sum_{r\geq 1} e^{-2\pi^2\sigma^2 r^2}\cos 2\pi r\ell$, within 1.5% of unity for $\sigma \geq \frac{1}{2}$ and within 10^{-8} for $\sigma \geq 1$; we restrict to $\sigma \geq \frac{1}{2}$.

For the reduced state (18) the Husimi function $Q(\theta,\ell) = \langle\theta,\ell|\,\rho_1\,|\theta,\ell\rangle$ is independent of θ,

$$Q(\ell) = \sum_m p_m\, g_\sigma(\ell - m), \tag{60}$$

the Schmidt weights smoothed by a Gaussian of width σ, and the Wehrl entropy [8, 28] is

$$S_W(\sigma) = -\int_0^{2\pi} \frac{\mathrm{d}\theta}{2\pi} \int \mathrm{d}\ell\; Q \ln Q = -\int \mathrm{d}\ell\; Q(\ell)\ln Q(\ell). \tag{61}$$

The same operation applied to the classical density gives $Q_{\mathrm{cl}} = w * g_\sigma$ and $S_W[w](\sigma)$, which tends to S_{cl} of (26) as $\sigma \to 0$. We write $S_W[p]$ when the distinction matters.

Bounds. Let $s(\ell) = \sum_m g_\sigma(\ell - m)$. Concavity of $\eta(x) = -x\ln x$ (Jensen's inequality with weights $g_\sigma(\ell-m)/s$) gives $\eta(Q) \geq \sum_m g_\sigma(\ell-m)\eta(p_m) - Q\ln s$, and integrating over ℓ,

$$S_q - \epsilon_\sigma \;\leq\; S_W \;\leq\; S_q + \tfrac{1}{2}\ln\left(2\pi e\sigma^2\right), \qquad \epsilon_\sigma \leq 2e^{-2\pi^2\sigma^2}(1 + O(e^{-2\pi^2\sigma^2})). \tag{62}$$

The lower bound is the Berezin–Lieb (Wehrl) inequality [29, 28, 30]; the upper bound follows from writing Q as the distribution of $X = M + Z$, with M distributed as p_m and Z an independent Gaussian of width σ, and $h(X) \leq H(M) + h(Z)$ [31].

7.5. Deep rotation: the Wehrl entropy in closed form

In deep rotation the Husimi entropy can be computed from the fringe structure (37). Over a stretch of many cells around j_0, w is nearly constant and $\Phi_j \simeq \Phi_0 + \kappa(j - j_0)$ with $\kappa = -2\beta$ by (38). By Poisson summation,

$$\sum_j e^{i\kappa j} g_\sigma(x - j) = \sum_{r\in\mathbb{Z}} e^{i(\kappa+2\pi r)x}\, e^{-\sigma^2(\kappa+2\pi r)^2/2}. \tag{63}$$

For $\sigma \gtrsim 1$ only the alias closest to zero survives, and

$$Q(x) \simeq w(x)\left[1 + A\cos\tilde{\Phi}(x)\right], \qquad A = e^{-\sigma^2 d(\kappa)^2/2}, \qquad d(\kappa) = \min_r |\kappa + 2\pi r|, \tag{64}$$

where $\tilde{\Phi}$ varies on the scale of a few cells. The smoothing does not remove the fringes; it reduces their contrast from 1 to A. Inserting (64) into (61) and averaging over the fast phase as in Sec. 4,

$$S_W \simeq S_{\mathrm{cl}} - \int \mathrm{d}x\; w(x)\, f\big(A(x)\big), \tag{65}$$

with the contrast function

$$f(A) = \big\langle (1 + A\cos\Phi)\ln(1 + A\cos\Phi)\big\rangle_\Phi = \ln\frac{1+\sqrt{1-A^2}}{2} + 1 - \sqrt{1-A^2}, \tag{66}$$

derived in Appendix A; $f(1) = 1 - \ln 2$ recovers (42). With $j = \gamma\cos\beta$ the arcsine weight is uniform in β, and $d(-2\beta) = \min(2\beta, 2\pi - 2\beta)$. Subtracting (42),

$$\boxed{S_W(\sigma) - S_q = (1 - \ln 2) - D(\sigma), \qquad D(\sigma) = \frac{2}{\pi}\int_0^{\pi/2} f\left(e^{-2\sigma^2\beta^2}\right)\,\mathrm{d}\beta} \tag{67}$$

Uniformity in γ. The approximations (39) and (64) both fail near the band edges, which is why $S_{\rm cl} - S_q$ approaches $1 - \ln 2$ only as $\gamma^{-1/3}$. But there $\beta \to 0$ or π, so $A \to 1$: the edge fringes are slow and are not touched by the smoothing. The integrand of $S_W - S_q$, $f(1) - f(A)$, vanishes where the approximations fail, and (67) holds without the $\gamma^{-1/3}$ correction.

Large σ. For $\sigma \gg 1$ the integrand is non-negligible only for $\beta \lesssim 1/\sigma$, the upper limit can be sent to infinity, and the series $f(A) = \sum_{n\geq 1}\binom{2n}{n}A^{2n}/[4^n 2n(2n-1)]$ (Appendix A) integrates term by term:

$$D(\sigma) \simeq \frac{c_1}{\sigma}, \qquad c_1 = \frac{1}{2\sqrt{\pi}}\sum_{n\geq 1}\frac{\binom{2n}{n}}{4^n\,2n(2n-1)\sqrt{n}} = 0.080171\ldots \tag{68}$$

The first term alone gives $1/8\sqrt{\pi} = 0.0705$; for $\sigma \geq 1$ the asymptotic form is accurate to better than 10^{-5}.

Numerical test. Table 1 compares (67) with the Wehrl entropy of the Bessel weights. At $\gamma = 10^4$ the agreement is within 0.001 for $1 \leq \sigma \leq 4$ (within 0.005 at $\gamma = 10^3$), and $S_W - S_q$ is the same at $\gamma = 10^3$ and 10^4 to within 0.004, although $S_{\rm cl} - S_q$ changes from 0.212 to 0.263 between them. At $\sigma = \frac{1}{2}$ the second alias, which matters near $\beta = \pi/2$ (fringes of period two cells), lowers the result by 0.003. At $\gamma = 30$ and 100 the larger widths begin to resolve the band itself and the Wehrl entropy rises above (67). Figure 6(a) shows the σ dependence. The equal-smoothing gap in the sixth column decomposes as $S_W[w] - S_W[p] = D(\sigma) - [(1-\ln 2) - (S_{\rm cl} - S_q)] + [S_W[w] - S_{\rm cl}]$: the residual fringe loss, minus the caustic correction to the unsmoothed deficit, plus the broadening of the classical band edges. At $\gamma = 10^4$ the three terms reproduce the column to 0.004.

Three consequences follow. First, the Husimi entropy of the quantum state does *not* converge to the classical entropy at fixed σ: as $\gamma \to \infty$,

$$S_W(\sigma) - S_{\rm cl} \;\longrightarrow\; -D(\sigma) \simeq -\frac{0.080}{\sigma}, \tag{69}$$

because a coherent state of fixed width only reduces the contrast of the fine fringes and does not remove it. Second, the classical value is reached only in the double limit $1 \ll \sigma \ll \gamma$, the upper bound being needed so that the smoothing does not broaden the band itself. Third, the remaining gap closes algebraically, as $1/\sigma$, not exponentially: fringes of all wavenumbers down to zero are present near the band edges, and a Gaussian of any width leaves the slowest of them intact.

7.6. Generic energies

At generic energies the local fringe wavenumber is $2\phi_k$, with ϕ_k the classical relative angle at momentum k (Sec. 4), and the same argument gives $S_W - S_q \simeq (1 - \ln 2) - D(\sigma; E_{\rm rel})$ with

$$D(\sigma; E_{\rm rel}) = \left\langle f\left(e^{-\sigma^2 d(2\phi)^2/2}\right)\right\rangle_{\rm orbit}, \tag{70}$$

Table 1: Wehrl entropy of the deep-rotation Schmidt weights $p_j = J_j^2(\gamma)$ against Eq. (67). $S_W[w]$ is the Wehrl entropy of the arcsine law smoothed with the same σ. $S_{\rm cl} = \ln(\pi\gamma/2)$.

γ	σ	$S_W[p] - S_q$	$(1 - \ln 2) - D(\sigma)$	$S_W[p] - S_{\rm cl}$	$S_W[w] - S_W[p]$	$S_{\rm cl} - S_q$
10^2	0.5	0.1406	0.1502	0.0453	0.0364	
	1	0.2218	0.2267	0.1265	−0.0110	0.0953
	2	0.2713	0.2668	0.1760	−0.0127	
	4	0.3300	0.2868	0.2347	−0.0040	
10^3	0.5	0.1468	0.1502	−0.0654	0.0912	
	1	0.2274	0.2267	0.0152	0.0213	0.2122
	2	0.2680	0.2668	0.0558	−0.0041	
	4	0.2916	0.2868	0.0795	−0.0064	
10^4	0.5	0.1468	0.1502	−0.1165	0.1247	
	1	0.2274	0.2267	−0.0359	0.0475	0.2633
	2	0.2675	0.2668	0.0042	0.0121	
	4	0.2877	0.2868	0.0244	−0.0013	

the average being over one period of the classical relative motion (the measure $w\,\mathrm{d}p$). Table 2 gives window averages at $\lambda I/\hbar^2 = 200$ and 800, computed with the classical density sampled uniformly in time along the orbit, which avoids the band-edge singularities. For $\sigma \le 1$, $S_W[p] - S_q$ agrees with (70) to within 0.03; for $\sigma = 2$ the bands, only 10–40 cells wide, are themselves broadened and the Wehrl entropy is larger than (70) predicts.

The equal-smoothing comparison is the fair one: quantum and classical distributions seen through the same coherent state. Its outcome is shown in Fig. 6(b) level by level at $\lambda I/\hbar^2 = 800$. For $\sigma = 1$ the two Wehrl entropies agree to within 0.04, and for $\sigma = 2$ to within 0.015, at every level with $-0.5 < E_{\rm rel}/\lambda < 0.8$ or $1.2 < E_{\rm rel}/\lambda < 3$, at both couplings; the window averages in Table 2 are below 0.03 and 0.01. The gap grows only close to the separatrix, where the classical period diverges. The unsmoothed deficit, by contrast, is about 0.2 in libration. The agreement is achieved, however, by moving *both* entropies: the last column of Table 2 shows that $S_W[w]$ exceeds the classical entropy $S_{\rm cl}$ by 0.1–0.7, depending on σ and on the width of the band.

7.7. What the phase-space route achieves

The three distributions — classical, Wigner, Husimi — give three different answers, and the rotor pair makes the relation between them exact.

1. The *Wigner* function of the reduced state is the marginal of the two-rotor Wigner function, and its entropy is the von Neumann entropy, identically (58). Using it in place of the classical distribution reproduces the quantum answer, including the deficit $1 - \ln 2$.
2. The *classical* distribution is the Wigner function with its chord terms averaged away *before* marginalisation. The constant $1 - \ln 2$ is the entropy of the chord terms, which survive the marginalisation as full-contrast fringes because conservation of J fixes the Schmidt basis.
3. The *Husimi* function removes the chord terms only partially. Its entropy exceeds S_q by $(1 - \ln 2) - D(\sigma)$, with D in closed form, and quantum and classical Husimi entropies

Table 2: Window averages over $0.2 \le E_{\rm rel}/\lambda \le 0.7$ (libration) and $1.5 \le E_{\rm rel}/\lambda \le 2.5$ (rotation), all levels of the sector $M = 0$. Column 3 is the unsmoothed comparison of Sec. 5.

$\frac{\lambda I}{\hbar^2}$	regime	$S_{\rm cl}^{\rm bin} - S_q$	σ	$S_W[p] - S_q$	$(1 - \ln 2) - D(\sigma; E_{\rm rel})$	$S_W[w] - S_W[p]$	$S_W[w] - S_{\rm cl}$
200	libration	0.180	0.5	0.193	0.204	0.025	0.147
			1	0.284	0.270	−0.007	0.207
			2	0.364	0.289	−0.002	0.292
200	rotation	0.012	0.5	0.146	0.146	−0.046	0.361
			1	0.273	0.223	−0.026	0.508
			2	0.459	0.264	−0.007	0.713
800	libration	0.208	0.5	0.202	0.204	0.031	0.104
			1	0.284	0.270	−0.008	0.147
			2	0.339	0.289	−0.003	0.207
800	rotation	0.051	0.5	0.144	0.146	−0.030	0.256
			1	0.246	0.223	−0.027	0.361
			2	0.374	0.264	−0.009	0.508

agree to $O(D(\sigma))$, i.e. to $\simeq 0.08/\sigma$ in deep rotation and, at generic energies away from the separatrix, to 0.04 for $\sigma = 1$ and 0.015 for $\sigma = 2$. But each Husimi entropy depends on σ, and no value of σ is singled out by the problem. The coherent-state width therefore takes over the role of the cell Δ in the oscillator problem of [1]: a parameter not fixed by the classical theory, whose choice decides the constant term. The cell itself, by contrast, remains fixed at h by the resolution of the identity.

One choice of σ suggests itself: the momentum width $\sigma_0 \propto (\lambda I/\hbar^2)^{1/4}$ of the pendulum's own ground state, Eq. (44), the analogue of the coherent states of an oscillator. With it the residual gap $D(\sigma_0) \propto (\lambda I/\hbar^2)^{-1/4}$ vanishes in the semiclassical limit, but only slowly, and the choice is a convention rather than a consequence of the dynamics.

8. Discussion

We have carried the programme of [1] from coupled oscillators to coupled rotors in the semiclassical regime, and extended it from the classical microcanonical distribution to the Wigner and Husimi distributions of the eigenstate.

The cell is fixed, and the equality is not exact. Compactness quantises the angular momentum, so the classical distribution is coarse-grained onto the integers that label the Schmidt weights and the comparison contains no free constant. With nothing to adjust, the quantum entropy lies below the classical one by $1 - \ln 2$ in the semiclassical limit. We derived this in deep rotation, where the Schmidt weights are $J_j^2(\gamma)$ and the classical weights the arcsine law, and confirmed it to $\gamma = 10^6$; at generic energies the same deficit is approached more slowly. The leading logarithms agree, as for the oscillators.

Why oscillators and rotors differ. The constant comes from the interference of the two classical branches compatible with one value of the observed momentum. For the rotors conservation of total angular momentum fixes the Schmidt basis, so the interference appears in the Schmidt weights; for the oscillators it is absorbed into the eigenvectors. In phase space, the

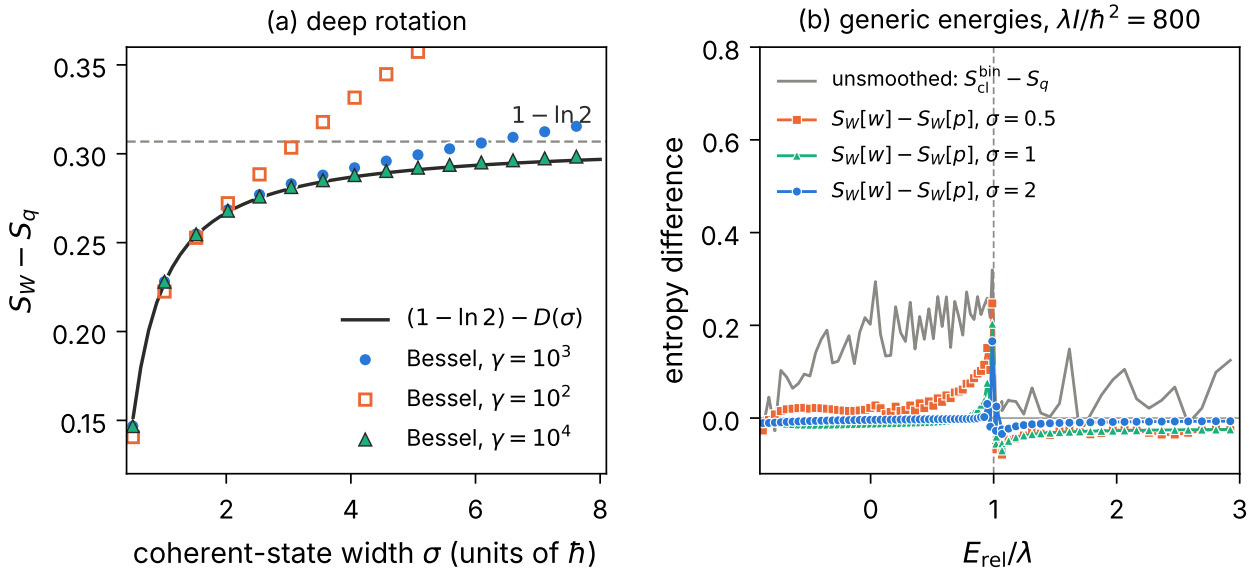


Figure 6: (a) Deep rotation: excess of the Wehrl entropy over the von Neumann entropy as a function of the coherent-state width, Eq. (67) (solid), against the Bessel weights at $\gamma = 10^2, 10^3, 10^4$ (symbols). At fixed σ the excess is independent of γ once $\sigma \ll \gamma$; the full deficit $1 - \ln 2$ (dashed) is recovered only as $\sigma \to \infty$. (b) Generic energies at $\lambda I/\hbar^2 = 800$: difference between the Wehrl entropies of the classical and the quantum momentum distributions for $\sigma = 0.5, 1, 2$, and the unsmoothed difference $S_{\rm cl}^{\rm bin} - S_q$ (grey). The dashed line marks the separatrix.

interference is the chord structure of the Wigner function, which marginalises into the Schmidt weights intact.

Phase-space distributions. Replacing the classical distribution by the Wigner function gives back the von Neumann entropy, identically; replacing it by the Husimi function gives an entropy between the quantum and classical ones, with a closed-form dependence on the coherent-state width. The Husimi route makes the quantum and classical distributions agree to $O(1/\sigma)$ when both are smoothed equally, but it does so by introducing a width that the problem does not fix. In this sense the rotor pair sharpens the conclusion of [1]: there is no classical calculation that reproduces the entanglement entropy including its constant term without one conventional parameter, be it a cell (oscillators), a coherent-state width (Husimi), or — when both are fixed, as here — an explicit correction $1 - \ln 2$.

The separatrix and deep libration. Both entropies are maximal near the separatrix, where the support of the classical momentum distribution has a cusp and where the excited-state quantum phase transition of the pendulum is located [16, 17, 18]. In deep libration the ground-state entropy is $\frac{1}{4}\ln(\lambda I/\hbar^2) + \frac{1}{2}\ln(\pi e/\sqrt{2})$ up to an anharmonic correction $-[8\sqrt{2\lambda I/\hbar^2}]^{-1}$, which we have verified to three digits.

Outlook. Coupling $N > 2$ rotors makes the relative motion non-integrable; whether the constant $1 - \ln 2$ and the Husimi law (67) survive chaos is the question most relevant to the high-energy setting that motivated [1]. The Wigner and Husimi functions are also natural tools for the dynamics of entanglement after a quench of λ across the separatrix, which the Mathieu machinery of Sec. 2 makes tractable.

Appendix A. Integrals

A.1. A logarithmic integral. Let $I_s = \int_0^\pi \ln\sin u\,\mathrm{d}u = 2\int_0^{\pi/2}\ln\sin u\,\mathrm{d}u = 2\int_0^{\pi/2}\ln\cos u\,\mathrm{d}u$. Adding the last two forms,

$$2I_s = 2\int_0^{\pi/2}\ln\frac{\sin 2u}{2}\,\mathrm{d}u = \int_0^\pi \ln\sin v\,\mathrm{d}v - \pi\ln 2 = I_s - \pi\ln 2, \tag{A.1}$$

so $I_s = -\pi\ln 2$; equivalently $\int_0^\pi \ln|\cos t|\,\mathrm{d}t = -\pi\ln 2$.

A.2. Entropy of the arcsine law. For $w(j) = 1/(\pi\sqrt{\gamma^2 - j^2})$ put $j = \gamma\cos\beta$; then $w\,\mathrm{d}j = \mathrm{d}\beta/\pi$ and

$$-\int w\ln w\,\mathrm{d}j = \ln\pi + \frac{1}{\pi}\int_0^\pi \ln(\gamma\sin\beta)\,\mathrm{d}\beta = \ln\frac{\pi\gamma}{2}. \tag{A.2}$$

A.3. The phase average. With $1+\cos\Phi = 2\cos^2(\Phi/2)$ and A.1, $\frac{1}{2\pi}\int_0^{2\pi}\ln(1+\cos\Phi)\mathrm{d}\Phi = \ln 2 - 2\ln 2 = -\ln 2$. Integrating by parts, $\frac{1}{2\pi}\int_0^{2\pi}\cos\Phi\ln(1+\cos\Phi)\mathrm{d}\Phi = \frac{1}{2\pi}\int_0^{2\pi}\frac{\sin^2\Phi}{1+\cos\Phi}\mathrm{d}\Phi = \frac{1}{2\pi}\int_0^{2\pi}(1-\cos\Phi)\mathrm{d}\Phi = 1$. The sum is $1-\ln 2$.

A.4. Band integrals. Rotation: with $u = \pi - 2t$, $E_{\mathrm{rel}} - \lambda\cos u = (E_{\mathrm{rel}}+\lambda)(1-k^2\sin^2 t)$, $k^2 = 2\lambda/(E_{\mathrm{rel}}+\lambda)$, and $B = \frac{\sqrt{I}}{2}\cdot\frac{2}{\sqrt{E_{\mathrm{rel}}+\lambda}}\int_0^{\pi/2}\mathrm{d}t\,(1-k^2\sin^2 t)^{-1/2}$. *Libration:* $u\in[u_0,\pi]$ with $\cos u_0 = E_{\mathrm{rel}}/\lambda$; with $v = \pi - u$ and $\sin(v/2) = \kappa\sin t$, $\kappa^2 = (E_{\mathrm{rel}}+\lambda)/2\lambda$, one has $E_{\mathrm{rel}} + \lambda\cos v = (E_{\mathrm{rel}}+\lambda)\cos^2 t$ and $\mathrm{d}v = 2\kappa\cos t\,\mathrm{d}t/\sqrt{1-\kappa^2\sin^2 t}$, so $B = \sqrt{I/2\lambda}\,\mathsf{K}(\kappa)$. The same substitutions parametrise the orbit uniformly in time, $\mathrm{d}t_{\mathrm{orbit}}\propto \mathrm{d}t/\sqrt{1-k^2\sin^2 t}$, which we use to sample the classical density in Sec. 7. Near the separatrix $\mathsf{K}(k) = \ln(4/k') + O(k'^2\ln k')$, $k' = \sqrt{1-k^2}$.

A.5. The contrast function $f(A)$. Let $F(A) = \langle\ln(1+A\cos\Phi)\rangle_\Phi$ and $G(A) = \langle\cos\Phi\ln(1+A\cos\Phi)\rangle_\Phi$ for $0\le A\le 1$, and $s = \sqrt{1-A^2}$. The standard result [32] $F(A) = \ln[(1+s)/2]$ generalises A.3. Integrating by parts as in A.3, $G = A\langle\sin^2\Phi/(1+A\cos\Phi)\rangle$. With $u = 1+A\cos\Phi$, $\sin^2\Phi = (A^2 - 1 + 2u - u^2)/A^2$, and using $\langle u^{-1}\rangle = 1/s$ and $\langle u\rangle = 1$,

$$G = \frac{1}{A}\left[(A^2-1)\frac{1}{s} + 2 - 1\right] = \frac{1-s}{A}, \qquad f(A) = F + AG = \ln\frac{1+s}{2} + 1 - s. \tag{A.3}$$

Alternatively, expanding $(1+y)\ln(1+y) = y + \sum_{k\ge 2}(-1)^k y^k/k(k-1)$ with $y = A\cos\Phi$ and $\langle\cos^{2n}\Phi\rangle = \binom{2n}{n}4^{-n}$,

$$f(A) = \sum_{n\ge 1}\frac{\binom{2n}{n}}{4^n\,2n(2n-1)}A^{2n} = \frac{A^2}{4} + \frac{A^4}{32} + \frac{A^6}{96} + \dots \tag{A.4}$$

With $\int_0^\infty e^{-4n\sigma^2\beta^2}\mathrm{d}\beta = \sqrt{\pi}/(4\sigma\sqrt{n})$ this gives (68).

Appendix B. Numerical methods and verification

All numerics use $\hbar = I = 1$ and the sector $M = 0$. The pendulum Hamiltonian is diagonalised in the two parity sectors (Sec. 2.6) with truncation $|k| \le N = 8\sqrt{\lambda} + 150$, the weight on the boundary states being below 10^{-6}. Classical cell weights (27) are computed by adaptive quadrature with the band-edge singularities supplied to the routine; for the arcsine law they are evaluated in closed form. Bessel weights are summed over $|j| \le \gamma + 40\gamma^{1/3} + 50$. The Wigner function is evaluated from (55) on 720 angles. Wehrl entropies of discrete weights are computed exactly on a sub-lattice of $K \ge 10/\sigma$ points per cell by convolution; Wehrl and differential entropies of the classical density are computed by sampling the orbit uniformly in time (A.4) with 2×10^5 points, histogramming at resolution 0.01 and convolving. Table B.3 collects the comparisons quoted in the text.

Table B.3: Analytic results against exact numerics ($\hbar = I = 1$). Mathieu checks at $q = 14.6$.

quantity	analytic	numerical
$M = 0$ lowest level vs $a_0(q)/4$	-5.4542890	-5.4542890
$M = 0$ third level vs $a_2(q)/4$	1.3368673	1.3368673
$M = 1$ lowest level vs $b_1(q)/4$	-5.4542787	-5.4542787
$M = 1$ second level vs $a_1(q)/4$	-1.9036729	-1.9036729
$N_{\rm lib}$ at $\lambda I/\hbar^2 = 800$ vs $(4/\pi)\sqrt{q}$	50.92958	51
B, $E_{\rm rel} = 300$, $\lambda = 100$, Eq. (24)	0.0927037	0.0927037
B, $E_{\rm rel} = 60$, $\lambda = 100$, Eq. (24)	0.1596085	0.1596085
$p_{j=0}$ at $n = 400$, $\gamma = 5$ vs $J_0^2(5)$	0.0315406	0.0314917
$p_{j=3}$ at $n = 400$, $\gamma = 5$ vs $J_3^2(5)$	0.1331018	0.1343519
arcsine entropy $\ln(\pi\gamma/2)$, $\gamma = 10^4$	9.6619231	9.6619231
$S_{\rm cl}^{\rm bin} - S_q$ at $\gamma = 10^6$ vs $1 - \ln 2$	0.3068528	0.2978385
S_0 at $\lambda I/\hbar^2 = 10240$, Eq. (49)	3.2067189	3.2067169
$\sum_\ell \int {\rm d}\varphi\, W_{\rm rel}$ ($\lambda = 200$, $E_{\rm rel}/\lambda = 1.54$)	1	1.000000000000
$\max_{\ell\in\mathbb{Z}} \lvert \int {\rm d}\varphi\, W_{\rm rel} - p_\ell \rvert$, property (iii)	0	3.8×10^{-17}
$\max_{\ell\in\mathbb{Z}+1/2} \lvert \int {\rm d}\varphi\, W_{\rm rel} \rvert$, property (iii)	0	3.0×10^{-17}
$\max_\varphi \lvert \sum_\ell W_{\rm rel} - \lvert u(\varphi)\rvert^2 \rvert$, property (ii)	0	3.0×10^{-13}
$\sigma D(\sigma)$ at $\sigma = 8$ vs c_1, Eq. (68)	0.080171	0.080171
$S_W - S_q$, $\gamma = 10^4$, $\sigma = 1$, Eq. (67)	0.2267	0.2274
$S_W - S_q$, $\gamma = 10^4$, $\sigma = 2$, Eq. (67)	0.2668	0.2675

The scripts that produce every figure and table are available from the authors.

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